

MD Tamim Rahman

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Work Authorization: Available (Sponsorship not required) | Open to Relocate

EDUCATION

University of Mississippi

Jan 2025 – Present

Bachelor of Science in Mechanical Engineering | GPA: 3.86 / 4.00

SKILLS

- **CAD / Simulation:** SolidWorks, AutoCAD (2D/3D), Creo, Ansys, Finite Element Analysis (FEA)
 - **Engineering Methods:** GD&T, DFM, DFA, Tolerance Stack-up, Engineering Drawings, BOM
 - **Manufacturing:** CNC Machining (Mill, Lathe), 3D Printing (FDM), Welding, Soldering, Fixture Assembly
 - **Programming & Tools:** MATLAB, Python, MS Office Suite (Word, Excel, PowerPoint, Outlook), Smartsheet
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EXPERIENCE

Undergraduate Research Assistant – Dept. of Mechanical Engineering, Univ. of Mississippi

Jan 2026 – Present

- Designed an FDM 3D-printed PLA heat sink with a graphene surface coating in SolidWorks using parametric geometry modeling, achieving a measured 64% improvement in thermal conductivity over baseline PLA across repeated trials.
- Optimized fin geometry and wall thickness across 6+ SolidWorks design iterations, increasing effective heat-dissipating surface area by ~30% while preserving 3D printability and applying DFM principles.
- Fabricated 8+ graphene-coated PLA prototypes via FDM 3D printing at a 90%+ print success rate, characterizing thermal performance through controlled bench-top testing to validate conductivity gains before each redesign cycle.
- Machined a metal heat sink reference part on manual mill and lathe, holding ± 0.01 cm dimensional tolerance and Ra 0.8 μ m surface finish per GD&T callouts for comparative thermal testing.
- Performed 10+ Ansys FEA thermal simulations to predict heat flux distribution, achieving results within ~7% of experimental measurements and refining material property assumptions across iterations.
- Analyzed 200+ thermal measurement data points in MATLAB and generated automated performance comparison plots, cutting manual analysis time by ~40% across prototype iterations.

Undergraduate Research Assistant – Dept. of BioMolecular Sciences, Univ. of Mississippi

Mar 2025 – Jan 2026

- Supported a Python-based behavioral detection system to quantify movement data from 50+ controlled trials, improving data validation and analysis efficiency by 20%.

Undergraduate Research Assistant – Dept. of Pharmacy & Drug Delivery, Univ. of Mississippi

Feb 2025 – Dec 2025

- Programmed and integrated automated bioprinting workflows for an NIH/FDA-funded research lab, improving lab throughput by 45%+ through optimized robotic process coordination (Wolkata bioprinting platform).

Summer Assistant – Dept. of Student Housing, Univ. of Mississippi

May 2025 – Jul 2025

- Coordinated residential transitions for incoming students by managing check-ins, room inspections, and orientation logistics under standardized housing procedures.
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PROJECTS

Thermal Simulation of Graphene-Coated PLA Heat Sink | SolidWorks Simulation, Ansys

Jan 2026 – Present

- Modeled steady-state thermal performance using SolidWorks Simulation with realistic heat flux boundary conditions and material thermal properties, validating the 64% conductivity improvement observed in physical testing.
- Iterated on fin geometry and graphene coating parameters through simulation runs to identify optimal design configuration, reducing peak temperature by isolating high-stress thermal zones prior to fabrication.

Multi-Part Mechanical Assembly Design (C-Clamp) | SolidWorks

Jan 2026 – Present

- Designed a fully constrained multi-component C-clamp assembly with GD&T-compliant tolerances and mate constraints to simulate real-world fit and function.
 - Produced manufacturing-ready engineering drawings with dimensional tolerances, surface finish callouts, and a complete bill of materials, applying DFM principles throughout.
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AWARDS

Chancellor's Honor Roll, University of Mississippi

Jun 2025

International Academic Excellence Award, University of Mississippi (\$7,500/year)

Sep 2024

International Non-Resident Academic Excellence Award, University of Mississippi (\$18,828/year)

Sep 2024

Higher Secondary Certificate Scholarship, Govt. of Bangladesh

2023